

Esther Manu

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EDUCATION

University of Cincinnati

Bachelor of Science in Electrical Engineering, GPA: 3.6

Cincinnati, OH

Expected Graduation: May 2029

- **Honors:** Dean's List (2x)
- **Relevant Coursework:** Network Analysis, Digital Design, Programming for ECE, Calculus I & II, Engineering Design, Semiconductor Physics, Differential Equations, Linear Algebra
- **Affiliations:** IEEE(UoC Chapter), NSBE, Colorstack, Association for Computing Machinery, Society of Women Engineers

TECHNICAL SKILLS

Languages: Python, C++, Arduino, Verilog HDL, Assembly, TypeScript

Hardware: Hardware Prototyping, Functional Testing, Measurement & Inspection, Process Documentation, Soldering

Software: Multisim, MATLAB, KiCad, LTspice, LabVIEW, GitHub

Tools & Concepts: Root Cause Analysis, Failure Mode and Effects Analysis, Statistical Data Analysis, Data Visualization

EXPERIENCE

Undergraduate Research Assistant

Feb. 2026 – Present

University of Cincinnati

Cincinnati, OH

- Conducted research on next-generation solar cells, integrating thin-film materials into prototype devices to improve power conversion efficiency (PCE).
- Performed functional testing and data collection across 6 prototype iterations using calibrated characterization equipment (I-V curve tracers, spectrophotometers), measuring PCE, fill factor, and open-circuit voltage.
- Analyzed and documented results to benchmark prototype performance against commercial standards, identifying quality gaps and driving a 24% improvement in efficiency.

Student Assistant

Jan 2025 – Nov. 2025

University of Cincinnati

Cincinnati, OH

- Independently set up, operated, and troubleshoot hardware and power conversion equipment during live STEM outreach demonstrations, ensuring safe and reliable operation.
- Diagnosed and resolved equipment malfunctions in real time to minimize demonstration downtime and maintain session quality.
- Directly contributed to university recruitment efforts, with 5 participants enrolling at UoC following outreach sessions.

Machine Learning Researcher

May 2025 – Oct. 2025

AI4ALL

Remote

- Built a Naïve Bayes text classification pipeline in Python to detect phishing emails, extracting word-frequency and weighted-term features from raw email text.
- Preprocessed and cleaned a dataset of 10,000+ emails through tokenization and stop-word removal, then applied feature selection to reduce dimensionality and improve model generalization.
- Tuned model hyperparameters (smoothing parameter, feature vocabulary size) via cross-validation, achieving 95% accuracy and 92% recall on test data.

PROJECTS

Lightweight Drone Frame Design | *AutoCAD, Python (OpenCV, Numpy, GeoPanda), MATLAB*

- Designed a modular agricultural drone airframe using AutoCAD to support precision farming through aerial crop monitoring.
- Conducted structural load analysis and design validation to improve flight stability and payload capacity.
- Integrated a camera and NDVI imaging system to capture crop health data, enabling farmers to detect plant stress and optimize field management.

DWIPS – Wearable Motion-Detection Safety System | *Sensors (MPU6050), Arduino, MATLAB, KiCad*

- Designed a solar-powered wearable safety device that uses motion sensors to detect falls and hazardous movement in real time.
- Integrated photovoltaic cells with the sensor and alert system to enable continuous operation without battery replacement.
- Tested and validated the prototype across multiple motion scenarios, tuning sensor sensitivity to minimize false positives while confirming reliable solar charging performance.

Image Processing System for Object Detection | *Python(OpenCV, Numpy, Pandas)*

- Built Python-based computer vision pipeline to automatically classify electronic components and flag defective units.
- Extracted visual features and applied statistical analysis to detect variations and trends across test batches, reducing manual inspection time.
- Validated classification performance against manually labeled test sets, improving component sorting accuracy and streamlining the testing workflow.

ACTIVITIES

Student Volunteer – Midwest Symposium on Circuits and Systems (MWSCAS): Supported conference registration, technical sessions, and the student-industry forum.

JPMC Data for Good Hackathon: Developed crime intelligence pipeline unifying 80K+ incidents from Chicago and Philadelphia; trained XGBoost on datasets and implemented transfer learning to deliver Delaware risk forecasts, aiding active criminal investigations.